



Can manual and automated trucks coexist efficiently? A data-driven evaluation of mixed operation mode in retrofitted container terminals

Chao Lei^a, Jiacheng Liu^a, Mei Sha^{a,*}, Mei Yang^b, Qi Wang^a, Zhongdai Wu^c, Bing Han^c

^a College of Transport and Communications, Shanghai Maritime University, Shanghai 201306, China

^b Shanghai International Port (Group) Co., Ltd, Shanghai 200086, China

^c Shanghai Ship and Shipping Research Institute Co., Ltd, Shanghai 200135, China

ARTICLE INFO

Keywords:

Container terminal retrofitting
Mixed operation mode
Machine learning
Operational efficiency

ABSTRACT

This study investigates the operational efficiency of container terminals utilizing a mixed operation mode, in which both manual and automated container trucks operate concurrently without physical separation. Drawing on real-world production data from a retrofitted container terminal, we employ a bagging-based machine learning model that integrates Random Forest, XGBoost, LightGBM, and Feed-Forward Neural Networks to accurately predict task durations under diverse operational scenarios. Shapley Additive Explanations are utilized to interpret the machine learning outcomes, identifying critical factors such as concurrent task volume, truck type configuration, and task type that significantly influence operational efficiency. Our findings indicate that mixed operation modes demonstrate superior performance for moderate workload scenarios, offering optimal efficiency through strategic configurations of truck types. This research contributes actionable insights into optimal equipment allocation and operational strategies, guiding decision-makers in effectively managing the transition toward automated container terminal operations.

1. Introduction

With the global demand for container transportation continuing to rise, the adoption of automation and intelligent operations has become imperative for container terminals. Despite the development of fully automated terminals, numerous traditional terminals worldwide still require retrofitting or partial reconstruction to integrate automation. As of 2022, only approximately 3 % of global container terminals had fully implemented automation (Knatz et al., 2022).

However, completely halting operations to undergo comprehensive automation upgrades is typically economically infeasible due to significant potential revenue losses. A practical and cost-effective alternative is to partially introduce automated equipment while maintaining existing manual infrastructures. For instance, the Hairun terminal at the Port of Xiamen, China, adopted a mixed operational mode, allowing automated container trucks (ACTs) and manual container trucks (MCTs) to operate concurrently within shared spaces without physical barriers. This operational approach was initiated in 2022 and has since consistently featured mixed truck modes (Container News Team, 2022).

Operations within a horizontal layout container terminal encompass multiple interconnected processes, including vessel arrivals and

berthing, loading and unloading containers using specialized cranes, container storage and sorting, customs clearance and inspection, and final transfer to land-based transportation. Fig. 1 illustrates the general workflow in horizontal-layout container terminals, highlighting the critical role of truck operations (① *Loading* and ② *Unloading*), which represent the primary focus of our study. Although ancillary processes (e.g., customs inspections) are included for context, our analysis concentrates specifically on the operational efficiency of mixed-mode truck operations.

We explicitly define “mixed operation mode” as the concurrent operation of ACTs and MCTs within the same operational spaces without physical barriers or lane segregation. While other terminal resources such as quay cranes or yard cranes undoubtedly influence operational efficiency, container trucks represent the critical bottleneck in horizontal transport operations, accounting for over 60 % of total task durations. Thus, this study specifically targets truck interactions as the central determinant of terminal operational efficiency under mixed operation modes.

Although existing research extensively covers purely automated or manual terminal operations (Song et al., 2024; Xu et al., 2024), limited research has addressed mixed-mode terminal environments—common

* Corresponding author.

E-mail address: meisha@shmtu.edu.cn (M. Sha).

<https://doi.org/10.1016/j.cie.2025.111539>

Received 25 March 2025; Received in revised form 14 July 2025; Accepted 17 September 2025

Available online 20 September 2025

0360-8352/© 2025 Published by Elsevier Ltd.

in ports transitioning toward automation—which provides superior intelligence, improved road utilization, and enhanced scheduling flexibility (Wu et al., 2022). Additionally, MCT operations introduce behavioral uncertainties—including variable acceleration, emergency braking, and inconsistent gap acceptance behaviors—that significantly affect traffic flow, operational efficiency, and safety (Ma et al., 2022). Yet, these effects have not been quantified or comprehensively analyzed in previous terminal operation studies. Consequently, investigating the operational dynamics of mixed-mode terminals is crucial for developing precise short-term operational strategies and effective long-term automation transition plans (Filom et al., 2022).

Urban transportation research has also investigated mixed operational scenarios involving automated and manual vehicles (Niroumand et al., 2023; Tajalli & Hajbabaie, 2022; Yang et al., 2025). However, these studies predominantly rely on simulated scenarios or theoretical assumptions due to the limited real-world coexistence of autonomous and manual vehicles in urban contexts. Container terminals, by contrast, offer unique contained environments with high-density vehicle interactions, fixed routing constraints, and fully integrated automated-manual vehicle operations already in place. This distinction enables our study to provide practically applicable, data-driven insights into real mixed-mode operations, filling a significant knowledge gap and providing valuable benchmarks for future optimization research.

Consequently, this study investigates the following research questions: (i) what operational factors critically affect terminal efficiency under mixed operational modes; (ii) how do interactions between ACTs and MCTs influence task durations across varying workload scenarios; and (iii) what practical operational insights and managerial implications can be derived from understanding these dynamics?

To address these critical managerial concerns, this study analyzes real-world operational data from container terminals employing a mixed operational mode. We develop a bagging-based machine learning model that integrates Random Forest (RF), Feed-Forward Neural Networks (FFNN), XGBoost, and Light Gradient Boosting Machines (LightGBM) to predict truck operational efficiency under diverse conditions and evaluate the impacts of different ACT and MCT configurations across various task scenarios. Additionally, we apply Shapley Additive Explanations

(SHAP) to identify and quantify the importance of key factors influencing operational efficiency. The results suggest that mixed operational modes are particularly effective for terminals with moderate daily workloads, where efficiency improvements exhibit phased rather than linear progression as ACT and MCT deployment scales increase. Strategic management of truck configurations enables terminals to optimize operational efficiency and enhance task completion rates.

This study contributes theoretically by explicitly identifying and quantifying critical factors that significantly affect operational efficiency under mixed operation modes. Unlike existing research primarily focused on fully automated or purely manual systems, our empirical analysis uniquely addresses the operational complexities introduced by simultaneous operations of ACTs and MCTs. By elucidating these unique interactions, our research provides a foundational empirical basis for developing advanced optimization frameworks, sophisticated simulation models, and robust decision-making methodologies tailored specifically to mixed-operation terminal environments.

The remainder of this paper is structured as follows. Section 2 reviews relevant literature on machine learning applications in container terminals and SHAP utilization across various fields. Section 3 outlines data processing procedures and preliminary analysis. Section 4 details the development and validation of the bagging model, employing SHAP for interpretative analysis. Section 5 presents an in-depth discussion of SHAP results and further evaluates the machine learning outcomes. Finally, Section 6 summarizes the study's primary conclusions and contributions.

2. Literature review

As critical logistics nodes, container terminals' interactions with external factors such as vessel arrival schedules and container throughput have garnered significant research interest. Li et al. (2024), for instance, incorporated causality analysis and transfer entropy methods to evaluate propagation risks associated with containership delays within container shipping networks.

Providing reliable analytical insights and recommendations is essential for terminal managers aiming to optimize short-term

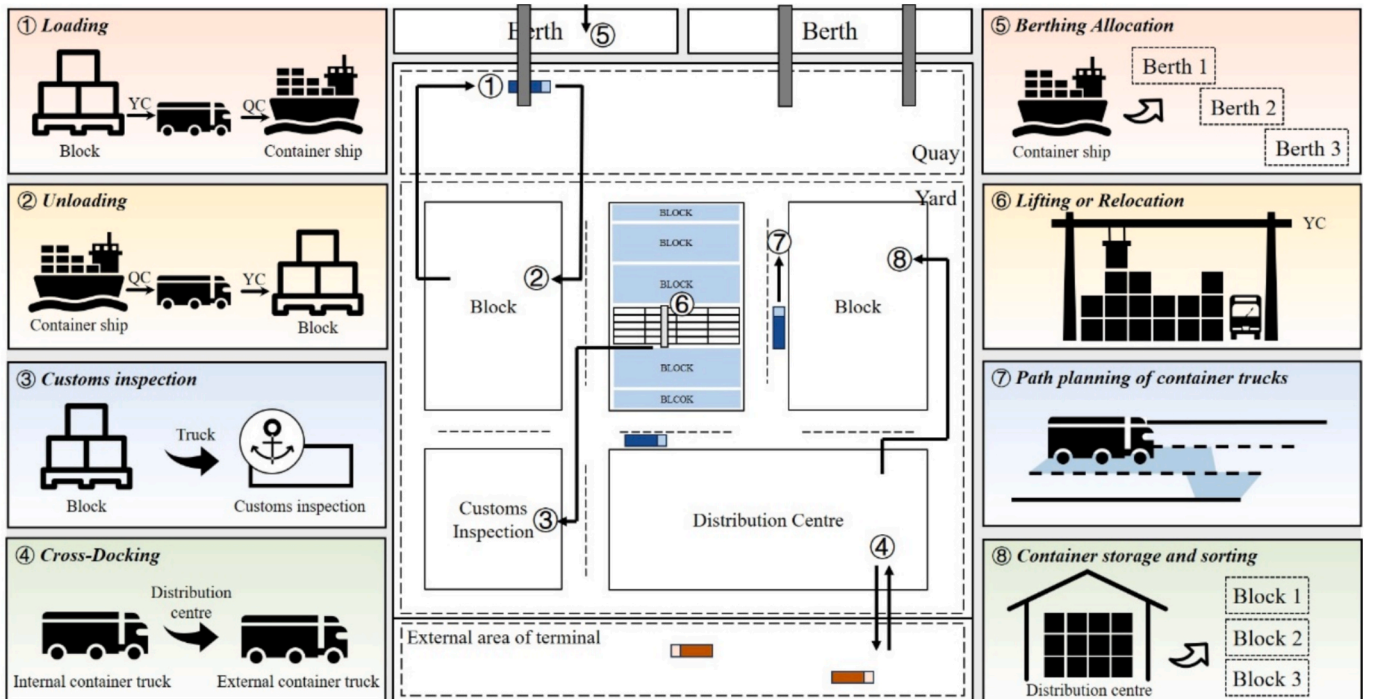


Fig. 1. Schematic diagram of primary operations in container terminals.

scheduling and formulate effective long-term strategies. Although vessel operators and truck drivers typically provide estimated arrival times in advance, these estimates often require updates due to unforeseen events (Pani et al., 2015). Recently, machine learning techniques have greatly enhanced predictive analytics, effectively capturing data complexities such as non-stationarity, volatility, seasonality, cyclicity, and randomness (Filom et al., 2022).

Several studies have utilized machine learning approaches to investigate diverse terminal-related challenges, including ship arrival prediction, pollution assessment, risk evaluation, and vessel movement optimization (Pani et al., 2015; Yu et al., 2018; Ozturk et al., 2019; Fabregat et al., 2021). For instance, Balster et al. (2020) applied machine learning models to predict arrival times for intermodal freight transport, accounting for processing times at inland terminals. Alvarillos et al. (2021) developed neural network and gradient boosting frameworks to forecast operational statuses of moored vessels based on ocean-meteorological conditions and vessel-specific characteristics, thereby optimizing operational locations and minimizing downtime-related costs. Similarly, Abreu et al. (2023) utilized various machine learning classification methods to predict vessel dwell times, defining operational time windows available at ports and identifying the most influential factors.

Research has also extensively explored container dwell times and throughput forecasting. Gao et al. (2019) analyzed five years of container traffic data to predict truck traffic volumes employing Long Short-Term Memory (LSTM) models, and compared the outcomes with Autoregressive Integrated Moving Average (ARIMA) and Back Propagation Neural Network (BPNN) models. Common predictive variables include container size and type, discharge dates, port of origin, and commodity classifications (Ping & Fei, 2013; Kourouniotti et al., 2016; Du et al., 2019; Milenković et al., 2021). Furthermore, following disruptions like the COVID-19 pandemic, terminal managers in Vietnam adopted neural network-based approaches for timely decision-making and maintaining port sustainability (Cuong et al., 2022).

Compared with studies addressing vessels and containers, research specifically focusing on container truck operations remains relatively limited. Existing literature predominantly addresses aspects such as path planning, scheduling, and equipment control, often focusing on external machinery associated with container transport volumes (Xie & Huynh, 2010; Luo & Huang, 2020; Adi et al., 2020; Skaf et al., 2021; Xu et al., 2021).

Moreover, mixed traffic operations involving manual and automated vehicles have been explored extensively within urban transportation research, primarily through simulation studies aimed at optimizing traffic flow, enhancing safety, and reducing congestion (Jiang & Shang, 2023; Niroumand et al., 2023; Tajalli & Hajbabaie, 2022; Wang et al., 2022). Nonetheless, these urban-focused studies rely on simplified assumptions and lack empirical validation since the widespread coexistence of automated and manual vehicles remains limited on public roads (Liu et al., 2024; Yang et al., 2025; Zhang et al., 2024). Container terminals, in contrast, provide distinctive real-world environments featuring mixed truck operations within confined spaces and fixed operational procedures. Despite the abundance of research on fully automated truck scheduling and path optimization in container terminals (Song et al., 2024; Tao et al., 2023; Xu et al., 2024), few studies have explicitly addressed the unique operational complexities and interaction dynamics arising from mixed manual and automated truck operations. Our study directly addresses this gap, utilizing actual operational data to quantitatively assess the impacts of mixed operational modes on terminal efficiency, thus significantly extending existing theoretical and practical knowledge.

This study employs machine learning methodologies to analyze the factors influencing container truck efficiency within mixed operation modes. However, solely relying on machine learning models for identifying key determinants of operational efficiency poses challenges due to their inherent complexity, non-linearity, and limited interpretability.

Abductive analytical techniques, such as SHAP, provide a robust method for clarifying complex model predictions. Introduced by (Lundberg & Lee, 2017), SHAP leverages Shapley values from cooperative game theory to quantify each feature's contribution to predictions systematically.

SHAP has successfully facilitated interpretability across various domains, including project management (Santos et al., 2023), wastewater treatment (Xie et al., 2024), medical diagnostics (Xu et al., 2024), urban planning (Yang et al., 2024), and industrial processes (Oliveira et al., 2024). The SHAP methodology comprises two primary steps: first, training the machine learning model; second, computing Shapley values by assessing differences in model predictions, considering subsets of features systematically. This calculation uses a representative background dataset to evaluate each feature's contribution across diverse scenarios. Integrating SHAP into our analysis enhances the interpretability and validity of machine learning predictions, offering a transparent understanding of factors influencing container truck operational efficiency within mixed-mode terminal environments.

3. Data processing and preliminary analysis

This section describes the case context and details the data utilized in our analysis. Using this data, we assess the operational efficiency of varying configurations of ACTs and MCTs, as well as interactions among different task types and operational scenarios.

3.1. Case setting

The container terminal analyzed in this study employs a horizontal layout featuring two operational bridges connecting quay areas and yard blocks, and comprises eight distinct yard zones containing a total of 72 blocks, as depicted in Fig. 2. The terminal utilizes a mixed truck fleet consisting of 140 MCTs and 30 ACTs. Operations follow explicit, pre-determined clockwise routing rules, ensuring standardized traffic flows, following established routes illustrated by blue arrows in Fig. 2. While ACTs are continuously available except during brief intervals for battery management, MCT availability is shift-dependent, reflecting real-world variability.

The terminal's operational scale, exceeding 2,000 TEU daily throughput, is indicative of significant operational complexity, representative of medium to large container terminals transitioning toward automated systems. While specific operational nuances (e.g., crane handling speeds, exact task prioritization schemes) naturally vary across terminals, our detailed operational characterization provides a sufficiently comprehensive context to ensure the conceptual transferability of our key insights to similar mixed-mode operational scenarios.

3.2. Data preprocess

The operational dataset utilized in this study originally comprised over 40,000 task records collected continuously from a container terminal over a complete one-month period (September 1–30, 2023). The dataset includes comprehensive details such as truck type (ACT or MCT), container size, vessel type, origin and destination yard locations, task dispatch and completion times, and segmented durations covering each distinct operational phase. By covering a full month of continuous operations, the dataset inherently captures both typical daily fluctuations and variations associated with peak vessel arrivals, including periods of high congestion as well as relatively lower-density operational intervals. After rigorous filtering (outlined below), approximately 20,000 high-quality, validated task records were retained for subsequent analysis.

Initial task records included various logging interruptions and incomplete entries due to typical system irregularities encountered in operational environments. To preserve analytical integrity, we implemented a rigorous data cleansing protocol, including: (i) records with any incomplete or missing data fields were completely excluded to avoid

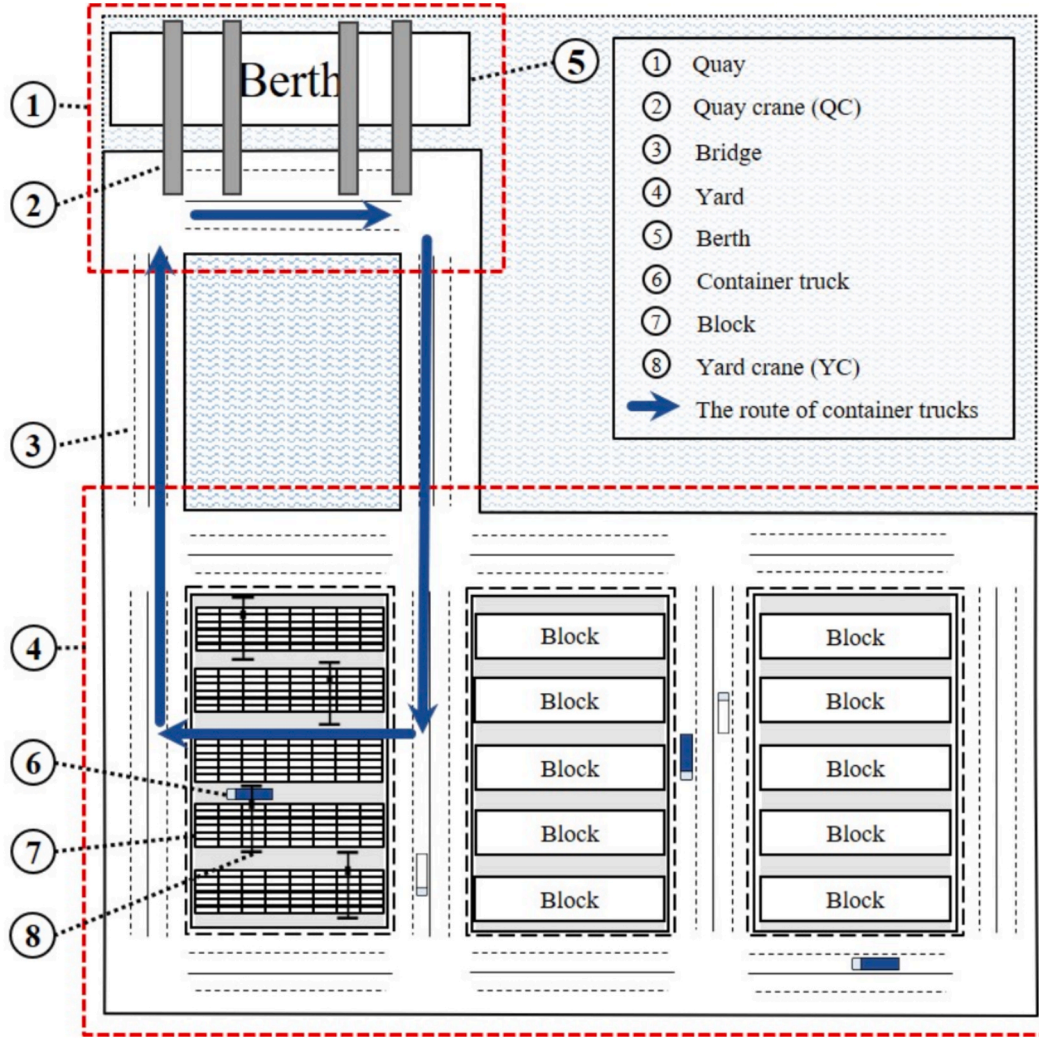


Fig. 2. Partial schematic layout of the container terminal.

introducing bias through imputation; (ii) entries with obviously implausible values, such as zero-second durations for critical operational segments (e.g., block operations, crane handling times), were removed, recognizing them as logging anomalies rather than genuine operational occurrences; and (iii) extreme outliers, such as durations significantly exceeding typical operational timeframes (over 99 % of validated tasks recorded durations under 4,800 s), were identified through statistical analysis and systematically excluded from further modeling. After this detailed quality-control process, approximately 20,000 robust and validated records remained for machine learning analysis. This stringent filtering ensures the reliability, validity, and representativeness of our dataset, significantly enhancing the accuracy and generalizability of derived insights.

To ensure the realism and accuracy of our predictive analyses, we strictly enforced adherence to actual operational constraints. Specifically, our dataset includes only tasks that fully complied with the terminal's established clockwise routing rules and task prioritization schedules. Any task records indicating deviations from prescribed operational procedures (e.g., incorrect routing direction, unscheduled truck dispatches) were identified during preprocessing and explicitly removed from the dataset. Consequently, the resulting machine learning analyses and interpretations inherently reflect realistic terminal operational scenarios.

Each operational task corresponds to a distinct loading or unloading event, involving container transfers from quay to yard or vice versa.

Task duration, D_{total} , measures the complete handling process from dispatch through final placement, which is defined as the sum of the block duration D_{block} , quay duration D_{quay} , road duration D_{road} , and bridge duration D_{bridge} , as expressed in Eq. (1). Each operational segment includes waiting time $D_{waiting}$ and active operation time $D_{operation}$, as specified by Eq. (2) and Eq. (3), respectively. Eq. (4) further partitions the total duration of the task into dispatch duration $D_{dispatch}$, defined as the interval from task assignment to the truck's arrival at the initial container position, and transfer duration $D_{transfer}$, covering the subsequent period from initial position arrival to task completion. The complete list of notations is shown in Table 1.

$$D_{total} = D_{block} + D_{quay} + D_{road} + D_{bridge} \quad (1)$$

$$D_{total} = D_{waiting} + D_{operation} \quad (2)$$

$$D_{block} = D_{waiting}^{block} + D_{operation}^{block} \quad (3)$$

$$D_{total} = D_{dispatch} + D_{transfer} \quad (4)$$

Fig. 3 illustrates selected task durations for different truck types under mixed-mode conditions. It is evident that multiple types of trucks frequently operate concurrently, and dispatch durations for MCTs often appear shorter than those for ACTs. We define concurrent tasks quantitatively as follows: a task j is considered concurrent with another task i if there is temporal overlap during their execution intervals. The

Table 1
List of notations.

Notation	Meaning
D_{total}	Total duration spent on a single operational task.
D_{block}	Duration spent on a single operational task in the block.
D_{quay}	Duration spent on a single operational task in the quay.
D_{road}	Duration spent on a single operational task on the road, excluding the internal block, bridge, and quay road.
D_{bridge}	Duration spent on a single operational task on the bridge.
D_{waiting}	Total waiting duration spent on a single operation task, caused by truck stagnation.
$D_{\text{operation}}$	Total operation duration spent on a single operation task, including container transferring and lifting duration.
D_{dispatch}	Duration spent on a single operational task from the task accepted location to
D_{transfer}	Duration spent on a single operational task from the location when the truck receives the dispatch order to the starting location of the task.
$D_{\text{block waiting}}$	The waiting duration for a single operation task in the block.
$D_{\text{block operation}}$	The operation duration for a single operation task in the block.
$N_j^{\text{concurrent}}$	The number of tasks operating concurrently with task j .
T_j	The total duration of task j .
T_i	The total duration of task i .

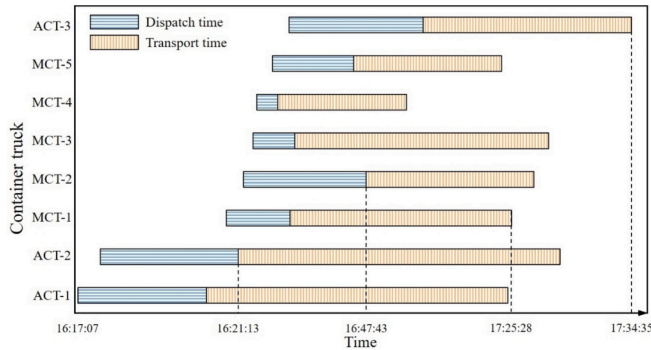


Fig. 3. Total task durations of loading operations under mixed-mode conditions.

concurrency for task j is calculated by the proportion of its total duration overlapping with task i , formally defined by Eq. (5):

$$N_j^{\text{concurrent}} = \sum_{i=1}^n (T_j \cap T_i) / T_j, j = 1, 2, \dots, n, i \neq j \quad (5)$$

here, $T_j \cap T_i$ indicates the overlapping interval between task j and task i , and T_j represents the full duration of task j . This explicit definition ensures consistent counting of concurrent tasks, accurately capturing true workload congestion rather than incidental overlaps.

Truck travel distances consist of the dispatch distance—from the dispatch information reception point to the task's initial position—and the subsequent transfer distance from the initial position to the endpoint. Dispatch distances are pre-recorded, while transfer distances are computed based on fixed operational routes, excluding minor adjustments for lane changes and turns.

3.3. Statistical analysis

Differences in operational efficiency between ACTs and MCTs, along with the impact of mixed-mode operations, are central managerial concerns. Statistical analyses indicate that MCTs typically achieve shorter total durations, dispatch durations, and transfer durations compared to ACTs. Furthermore, performance metrics for MCTs in entirely manual environments outperform those observed in mixed-mode operations. This performance disparity likely arises from interactions with ACTs.

Fig. 4 illustrates the distribution of task durations under various operational scenarios involving ACTs and MCTs. The categories defined are as follows: **ALL** refers to the complete set of observed tasks; **MIX** includes tasks performed in a mixed-mode scenario with both truck types operating concurrently; **MCT** and **ACT** denote tasks executed exclusively by either MCTs or ACTs, respectively; **M–MCT** and **M–ACT** represent tasks carried out specifically by MCTs or ACTs within mixed-mode conditions; finally, **P–MCTs** and **P–ACTs** indicate tasks exclusively performed by MCTs or ACTs within a purely manual or purely automated environment, without interference from the other truck type.

Safety protocols mandate that ACTs possess the lowest traffic priority, obliging them to yield and halt upon obstacle detection (e.g., pedestrians, vehicles, or equipment). In fully automated scenarios, centralized dispatching largely mitigates operational conflicts. However, such stringent safety measures considerably extend ACT task durations in mixed operations. Moreover, regulated maximum speeds and restricted overtaking further constrain ACT operations. MCT drivers frequently experience car-following scenarios with slower ACTs, hindering their speed and operational efficiency. Conversely, unrestricted ACTs follow optimal acceleration patterns but must decelerate or stop frequently when following unpredictable MCT behavior, exacerbating delays and inefficiencies. Consequently, task durations for **M–MCT** often exceed those of **ACT**, though this discrepancy may not be explicitly evident in raw data.

Beyond mixed-mode impacts and speed constraints, divergent operational objectives between ACTs and MCTs contribute significantly to efficiency differences. ACTs aim for precise adherence to scheduled task completion times to avoid queuing under cranes, necessitating slower, intermittent travel. In contrast, MCT drivers endeavor to arrive as early as possible to complete more tasks and maximize personal earnings.

We employed Spearman's rank correlation coefficient to evaluate associations among candidate features, due to its robustness against non-normality and data outliers typical in operational datasets. Spearman's correlation analysis, a nonparametric measure of statistical association that assesses the strength and direction of a monotonic relationship between two variables, was employed, with multicollinearity assessed by variance inflation factors, ensuring robust feature selection (Fig. 5). Rather than relying on the raw data values, Spearman's method converts each variable's observations into ranks and then computes the Pearson correlation coefficient on these ranks. The coefficient, denoted by ρ as in Eq. (6), ranges from -1 (perfect negative monotonic correlation) to $+1$ (perfect positive monotonic correlation), with 0 indicating no monotonic association.

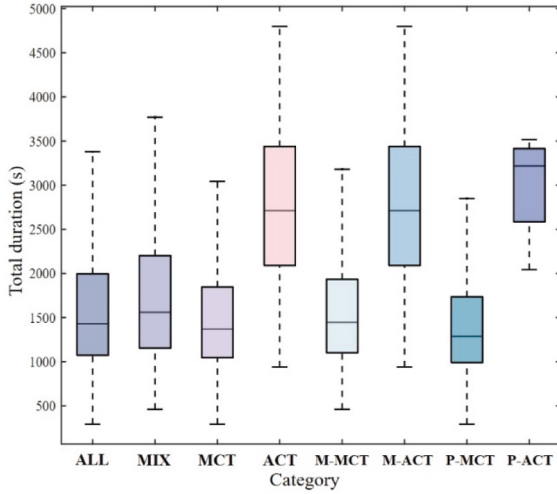
$$\rho = 1 - \frac{6 \sum d_i^2}{n(n^2 - 1)}, \quad (6)$$

where n is the number of paired observations. d_i is the difference between the ranks of the i -th pair.

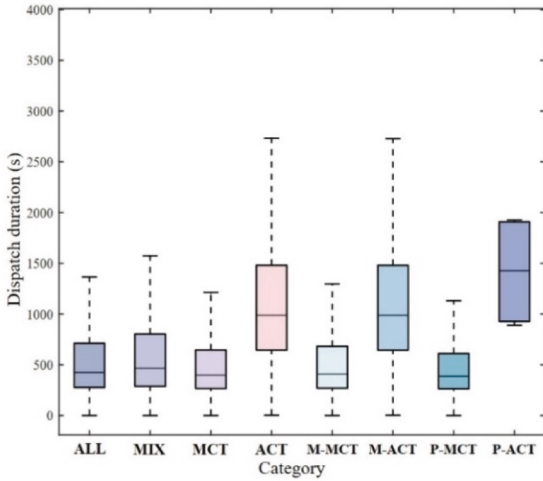
Operational factors such as task volume, truck numbers, truck types, human intervention, and container relocation also affect task durations. Spearman correlation analysis, a non-parametric approach suitable for non-normally distributed data, is applied (Fig. 5). Larger circles represent stronger correlations. Label definitions are provided in Table 2. Categorical encodings (e.g., T1, T2, T3) facilitate efficient modeling but may inadvertently suggest hierarchical relationships, potentially biasing results.

Operational influences broadly encompass three categories:

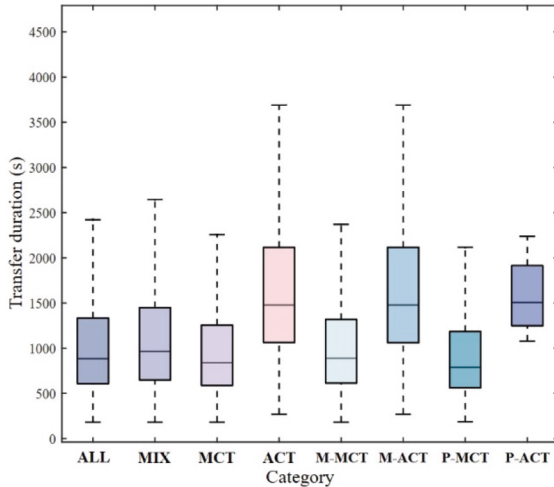
- 1. Task Number Influence:** Increased concurrent tasks and larger vessels contribute to congestion and longer waiting times.
- 2. Equipment Type Influence:** Variations among truck and crane types, along with human interventions, affect operational efficiency.
- 3. Task Operation Influence:** Container relocations and exchanges introduce additional waiting periods and extended distances, prolonging task durations.



(a) Total duration under different configurations



(b) Dispatch duration under different configurations



(c) Transfer duration under different configurations

Fig. 4. Task durations under different truck operational configurations.

4. Machine learning approach

Traditional statistical approaches primarily handle linear, low-dimensional datasets, heavily relying on predefined assumptions and manual feature engineering. In contrast, machine learning algorithms inherently accommodate complex, nonlinear interactions among operational variables, making them particularly suited to analyzing the multifaceted and dynamic nature of container terminal operations. Employing these methods in our study not only enhances prediction accuracy but also provides transparent operational insights critical for effective terminal management.

4.1. Feature selection

Feature selection significantly influences the performance of machine learning models by reducing model complexity, avoiding overfitting, and accelerating training. While redundant features may introduce noise, increase computational complexity, and degrade model accuracy, redundancy itself does not necessarily indicate irrelevance, as redundant variables may still provide useful information for the target prediction (Li et al., 2024).

Several variables listed in Table 2, such as “Task number,” “ACT number,” “MCT number,” “Loading number,” and “Unloading number,” exhibit redundancy due to their strong correlation and derivation from the total task count. Specifically, since the sum of ACT and MCT tasks and loading and unloading tasks equal the total number of tasks, these variables effectively act as proxies for overall task quantity. Moreover, ship type primarily influences the operational workload since larger vessels typically generate higher task volumes compared to smaller barges.

Certain variables remain essential for analysis despite apparent overlap. For example, although both “Task distance” and “Intersection number” reflect aspects of truck travel, only “Intersection number” captures the impact of frequent stops at traffic signals. Additionally, YC relocations may introduce complex nonlinear impacts on task durations. Although proactive relocations during idle periods can improve efficiency, incomplete relocations during peak periods lead to extended waiting times. Human intervention is also exclusive to MCT operations, significantly affecting efficiency.

Features exhibiting correlation coefficients greater than 0.6 (absolute value) were considered highly collinear and consequently aggregated into composite variables or removed to prevent multicollinearity issues. For example, T9 and T10 are combined into a single feature F8, representing the overall travel distance of the task; T13 and T14 are merged into F11, representing the number of concurrent tasks operated by different types of trucks; T15 and T16 are aggregated into F12, representing the number of concurrent tasks of the same type.

Based on these considerations, selected features for machine learning modeling are summarized in Table 3.

4.2. Model training and testing

Given the randomness of container ship arrivals and the absence of clear seasonal patterns in truck operations, traditional time-series approaches such as LSTM, RNN, and ARIMA are not suitable. Instead, widely adopted machine learning regression methods in container terminal studies fall into three primary categories: neural networks (Cuong et al., 2022; Milenković et al., 2021; Gökkuş et al., 2017), gradient boosting algorithms (Alvarellos et al., 2021; Fabregat et al., 2021; Stepec et al., 2020), and decision trees (Kim et al., 2021; Ozturk et al., 2019; Pani et al., 2015). Neural networks effectively capture nonlinear relationships. Gradient boosting algorithms like XGBoost and LightGBM iteratively minimize prediction errors, whereas decision tree methods such as Regression Trees (RT) and RF recursively partition feature spaces for robust predictions.

To accurately model nonlinear operational relationships, we

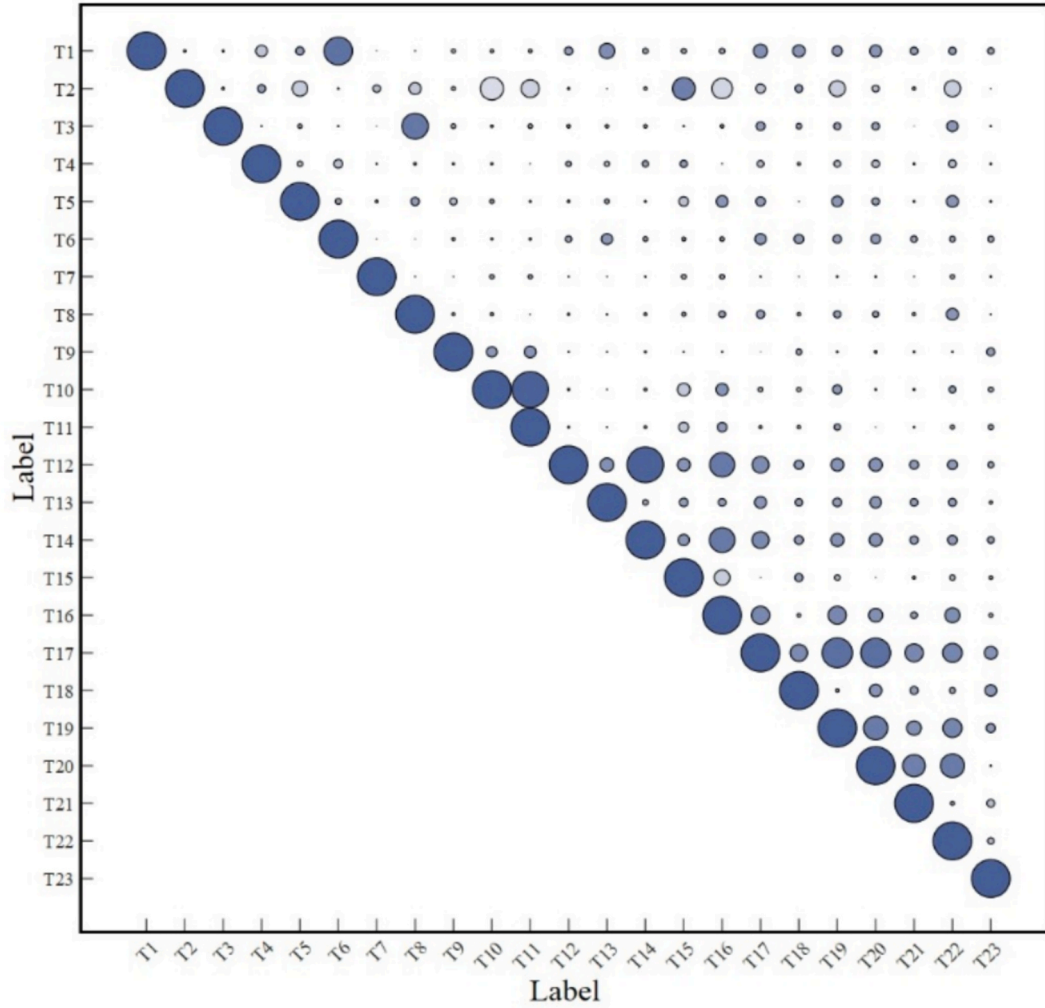


Fig. 5. Spearman correlation analysis results of operational factors.

employed machine learning techniques including RF, XGBoost, LightGBM, and FFNN within a Bootstrap Aggregating (Bagging) ensemble framework (Breiman, 1996). This ensemble approach ensures robust predictions and enhanced generalization by averaging multiple independently trained models. Such models have demonstrated proven effectiveness in capturing complex dependencies in port operations, as supported by recent literature (Filom et al., 2022). Although traditional, our machine learning framework is specifically tailored to address mixed operational dynamics. These methods cover a broad spectrum of tree-based and neural approaches suited to capturing the complex, nonlinear interactions present in mixed-mode terminal operations. By combining robust feature engineering, ensemble modeling, and SHAP interpretability, we uniquely contribute practical insights into mixed terminal operations. The Bagging procedure involves:

Generating bootstrap subsets from the original training dataset through random sampling with replacement.

Training base models (RF, XGBoost, LightGBM, and FFNN) on each subset, selected due to superior performance relative to BPNN.

Averaging predictions across all base models to obtain the final prediction.

This study evaluates and compares RT, RF, XGBoost, LightGBM, FFNN, BPNN, Genetic Algorithm optimized BPNN (GABP), and the Bagging ensemble method. Computations were executed on a PC equipped with an Intel Core i9-13900 CPU (2.50 GHz) and 128 GB of

RAM.

We assess model performance using Mean Squared Error (MSE, Eq. (7)), Mean Absolute Error (MAE, Eq. (8)), Explained Variance Score (EVS, Eq. (9)), and the coefficient of determination (R^2 , Eq. (10)). Lower MSE and MAE, along with EVS and R^2 closer to 1, indicate superior predictive performance.

$$MSE = \sum_{i=1}^n (y_i - \hat{y}_i)^2 / n, \quad (7)$$

$$MAE = \sum_{i=1}^n |y_i - \hat{y}_i| / n, \quad (8)$$

$$EVS = 1 - \frac{Var(y_i - \hat{y}_i)}{Var(y_i)}, \quad (9)$$

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2}, \quad (10)$$

where y_i is the actual value, $\hat{y}_i$ is the predicted value, $\bar{y}$ is the average of each actual value y_i , and n is the number of samples, and $Var()$ denotes variance.

Table 4 and Fig. 6 summarize model performance. All models except RT achieve comparable outcomes, with Bagging demonstrating the highest overall predictive accuracy. Although GABP generally shows

Table 2
Descriptions of data labels.

Code	Label	Description
T1	Container truck types	Type of container trucks: Manual Container Trucks (MCTs = 0) or Automated Container Trucks (ACTs = 1).
T2	Task types	The types of tasks, including loading(0) and unloading (1).
T3	Container types	Container types: 40-foot container (0), 20-foot container (1), or double 20-foot containers (2).
T4	Ship types	Type of container ships: Barges (0) or large ships (1).
T5	YC types	Types of yard cranes: Manual (0), Intelligent (manual with intelligent sensors, (1), or Automated (2).
T6	Human intervention	Human intervention requirement: Yes (1) or No (0).
T7	Container exchange	Container exchange between two trucks without yard stacking: Successful exchange (1), Failed exchange (2), or No exchange (0).
T8	Container relocation	Container relocation within the block: Yes (1) or No (0).
T9	Dispatch distance	Distance traveled from truck dispatch location to the initial task position.
T10	Transfer distance	Distance traveled from initial task position to the task completion location.
T11	Intersection number	Number of intersections encountered during truck travel.
T12	Task number	Number of concurrently performed tasks.
T13	ACT number	Number of ACTs operating tasks.
T14	MCT number	Number of MCTs operating tasks.
T15	Unloading number	Number of unloading tasks.
T16	Loading number	Number of loading tasks.
T17	Task duration	Total duration of a task.
T18	Dispatch duration	Time from receiving dispatch instructions to arrival at the initial task position.
T19	Transfer duration	Time from arrival at the initial task position to task completion.
T20	Waiting time	Total waiting time experienced during a task.
T21	Quay duration	Total duration spent at the quay for a task.
T22	Block duration	Total duration spent in the block for a task.
T23	Road duration	Total duration spent on road travel for a task.

Table 3
Selected features used in machine learning models.

Feature	Description
F1	Container truck type (MCT or ACT)
F2	Task type (loading or unloading)
F3	Container type (40-foot, 20-foot, or double 20-foot containers)
F4	Yard crane type (manual, intelligent, or automated)
F5	Human intervention requirement (yes or no)
F6	Container exchange status (successful, failed, or none)
F7	Container relocation within the block (yes or no)
F8	Task travel distance
F9	Number of intersections encountered
F10	Number of concurrent tasks
F11	Number of concurrent tasks operated by different types of trucks
F12	Number of concurrent tasks of the same type

Table 4
Predictive performance comparison of machine learning models.

Method	MSE	MAE	EVS	R^2
RT	574711.38	554.72	0.21	0.21
RF	286125.67	409.34	0.61	0.61
XBGoost	309599.85	423.51	0.57	0.57
Light GBM	290591.67	414.58	0.60	0.60
FFNN	294386.88	414.12	0.60	0.59
BPNN	304563.89	425.03	0.58	0.58
GABP	305493.24	425.30	0.56	0.56
Bagging	280851.02	407.68	0.62	0.62

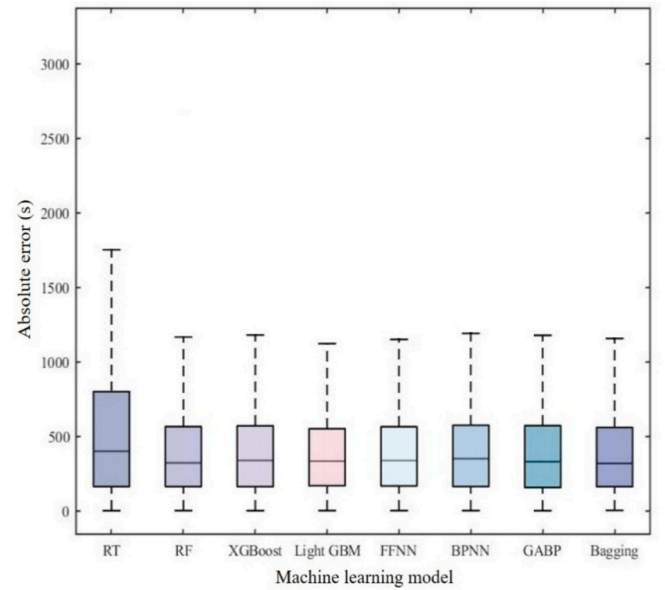


Fig. 6. Errors of various machine learning methods for predicting task durations.

smaller errors, its performance deteriorates due to frequent outliers, resulting in higher MSE and MAE than BPNN.

4.3. Results and discussions

As indicated by feature F2, both SHAP analysis and statistical results suggest that loading tasks generally require longer durations than unloading tasks. Terminal managers frequently question whether this difference persists when loading and unloading tasks share the same initial and target positions, and if so, what underlying factors contribute to the discrepancy. Because isolating such tasks in practice is challenging, we utilized the bagging model to simulate production scenarios and provide insights unattainable through statistical analysis alone.

We randomly selected 400 data points from the production dataset to evaluate loading and unloading task durations. For each loading task, we replaced its F2 feature value with the corresponding unloading task value, and vice versa. The bagging model's prediction results, presented in Fig. 7, do not indicate a definitive trend regarding whether loading or unloading tasks inherently require more time.

These outcomes suggest that the typically longer loading durations result primarily from additional operations and waiting times, consistent with the SHAP analysis findings. Therefore, to improve loading task efficiency, terminals should optimize scheduling strategies, utilize advanced algorithms to reduce travel distances, enhance container handling processes, logically sequence loading tasks to minimize unnecessary transfers, and provide targeted operator training. Table 5 presents the initial values of the relevant features.

4.3.1. Efficiency analysis for mixed mode

By varying features F1, F4, and F11 while keeping other parameters constant, we analyze operational efficiency under diverse configurations of ACTs and MCTs. Fig. 8 illustrates that task durations increase with the number of concurrent tasks operated by different types of trucks in operation, irrespective of whether the YCs are automated, intelligent, or manual. Notably, MCTs consistently exhibit higher efficiency than ACTs, supporting previous statistical findings. This efficiency advantage is most pronounced when fewer concurrent tasks are operated by different types of trucks but diminishes as truck diversity increases.

This trend occurs because, with fewer different trucks, interactions between ACTs and MCTs are minimal, allowing MCTs to fully utilize

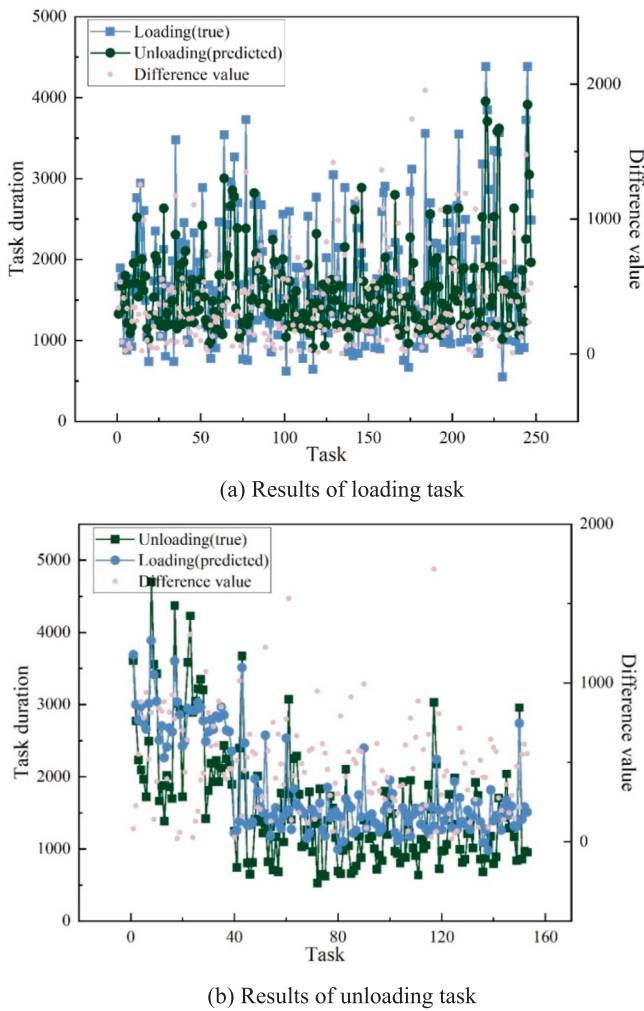


Fig. 7. Comparison results of loading (a) and unloading (b) tasks.

Table 5

Initial values of features used for the machine learning model analysis.

Feature	Value	Feature	Value
F1	0	F7	0
F2	0	F8	1500
F3	0	F9	3
F4	0	F10	20
F5	0	F11	40
F6	0	F12	40

their speed and flexibility. However, as truck diversity grows, ACTs impose safety-related operational constraints, causing frequent stops and slower speeds, reducing the efficiency of MCTs. Under conditions with numerous MCTs, ACTs are particularly disadvantaged, frequently needing abrupt stops to maintain safe distances.

To systematically analyze the impact of mixed truck operations on efficiency, we conducted experiments by fixing the number of ACTs at 20 and adjusting the total concurrent tasks to 40, 60, 80, 100, and 120, as shown in Fig. 9, corresponding to ACT:MCT ratios of 1:1, 1:2, 1:3, 1:4, and 1:5, respectively. Task durations under these varying ratios were analyzed to quantify how truck configuration impacts operational efficiency.

Our results clearly identify thresholds and critical operational ranges in the interaction between ACTs and MCTs, offering empirical benchmarks crucial for developing optimization models. Specifically, our findings indicate that operational efficiency significantly degrades when

the number of concurrent MCT tasks surpasses certain thresholds (i.e., 50 tasks). This empirically-derived threshold provides essential insights that future theoretical studies can incorporate into constraints and objectives of optimization models aimed at optimal resource allocation, scheduling, and task prioritization under mixed-mode operations.

When all other features are fixed, considering a loading task, evaluate five loading-to-unloading ratios (i.e., 2:8, 3:7, 4:6, 5:5, and 6:4) to understand how task composition interacts with the concurrent workload, as shown in Fig. 10. At a fixed total task volume, when the loading-to-unloading ratio changes from 2:8 to 3:7 to 4:6, the task duration remains approximately the same, indicating similar levels of interference from the same type of tasks across these three scenarios. However, once the ratio reaches 5:5, the task duration increases markedly, reflecting a significant rise in the same type of congestion. At 6:4, task duration increases further, although the magnitude of this increase is less pronounced than the jump observed between the 4:6 and 5:5 scenarios.

These findings suggest that when the same type of tasks begins to constitute half or more of the workload, interference effects intensify rapidly, but beyond that threshold, additional same-type tasks produce diminishing marginal delays. Moreover, between 30 and 40 tasks, durations stabilize, possibly decreasing slightly. However, durations sharply increase beyond 40 tasks, likely due to terminal congestion and machine learning model limitations arising from sparse high-volume training data.

In summary, task composition, particularly the ratio of loading to unloading tasks, significantly influences operational efficiency. Maintaining a balanced task ratio mitigates congestion effects, a key insight that can guide future theoretical models in defining operational constraints and scheduling strategies. And during off-peak periods the total number of concurrent tasks should be kept below approximately 40 to prevent the nonlinear surge in task durations observed when the same type of tasks exceeds this threshold. These empirical observations provide critical boundary conditions and operational benchmarks, thereby facilitating the development of sophisticated simulation studies and optimization frameworks that accurately represent realistic terminal operations.

4.3.2. Comparative analysis of automated and mixed mode

Given its unique balance of flexibility and stability, mixed operation mode could emerge as a promising operational approach. Decision-makers, however, remain interested in understanding performance differences between mixed and fully automated modes.

Fig. 11 compares task durations for mixed and fully automated modes as the number of concurrent tasks of the same type increases. As task numbers rise, the efficiency gap between mixed mode and fully automated mode becomes more pronounced. Fully automated mode generally yields shorter task durations, underscoring its efficiency advantages when handling large volumes of the same operation tasks. For example, at 30 tasks, the difference between the two modes is minimal. At 40 tasks, mixed mode durations increase significantly, while fully automated mode remains relatively stable. By the time 50 tasks are reached, the mixed mode's task duration spikes, whereas the fully automated mode remains low, emphasizing the latter's benefits.

However, although the automated mode consistently maintains lower durations when the number of concurrent tasks of the same type is high, the mixed mode outperforms the automated mode at fewer than 15 tasks. This occurs because minimal interactions in low-volume scenarios allow manual operations greater flexibility and efficiency. Beyond 50 tasks, fully automated operations demonstrate superior efficiency.

To deepen this analysis, we randomly selected 400 tasks from actual production data, replacing ACTs and YCs with fully automated equipment while maintaining other parameters unchanged. Fig. 12 illustrates that, although task durations in automated mode are somewhat higher, they exhibit greater stability compared to mixed mode.

Given that these automated-mode simulations retained mixed-mode

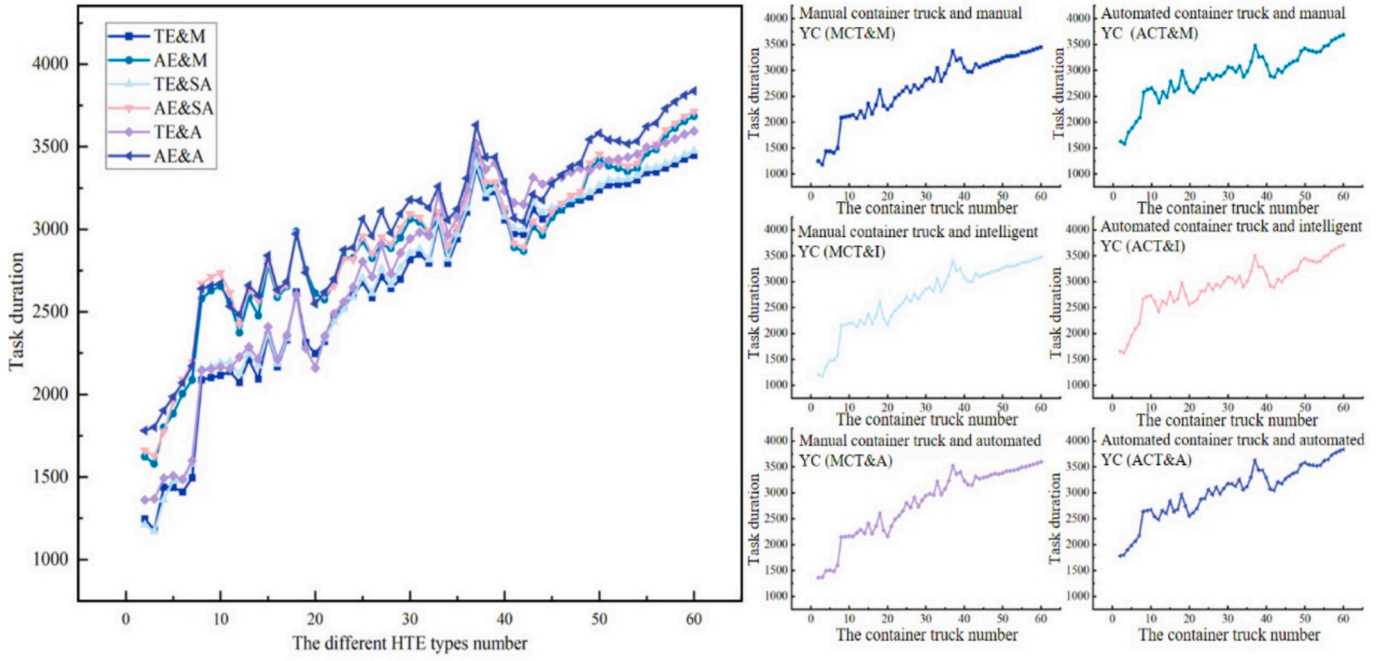


Fig. 8. Task durations for varying truck configurations.

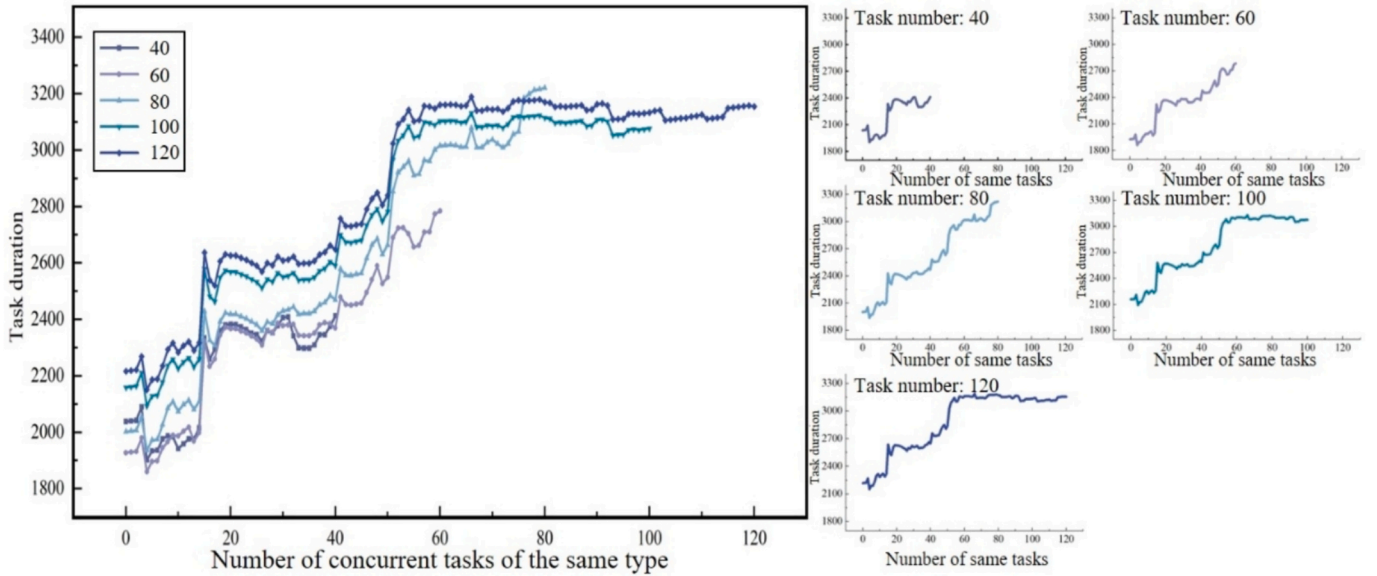


Fig. 9. Task durations at varying task volumes and truck configurations.

parameters (such as speed and safety distances), actual fully automated mode performance would likely improve further. Thus, fully automated terminals provide consistent stability advantages, whereas mixed modes offer faster completion in lower-volume and flexible operational contexts.

Considering Fig. 11 and Fig. 12 collectively, although automation provides greater operational stability, the mixed mode typically offers superior performance in realistic scenarios, particularly at moderate task volumes or amid unexpected operational changes. Since relatively few instances involve more than 50 tasks in actual production, the mixed mode likely outperforms the fully automated mode most of the time.

5. SHAP interpretations

The SHAP (Shapley Additive Explanations) method employs Shapley values derived from cooperative game theory. By considering features as cooperative contributors to model predictions, SHAP evaluates all possible subsets and permutations of features, thus providing a unified measure of feature importance.

Fig. 13 outlines the two-step process for computing Shapley values. The first step involves training the bagging model described in Section 4.2. In the second step, Shapley values for each feature are calculated based on their contributions to model predictions. This calculation includes selecting a background dataset representing the overall data distribution, evaluating all subsets of features, computing the

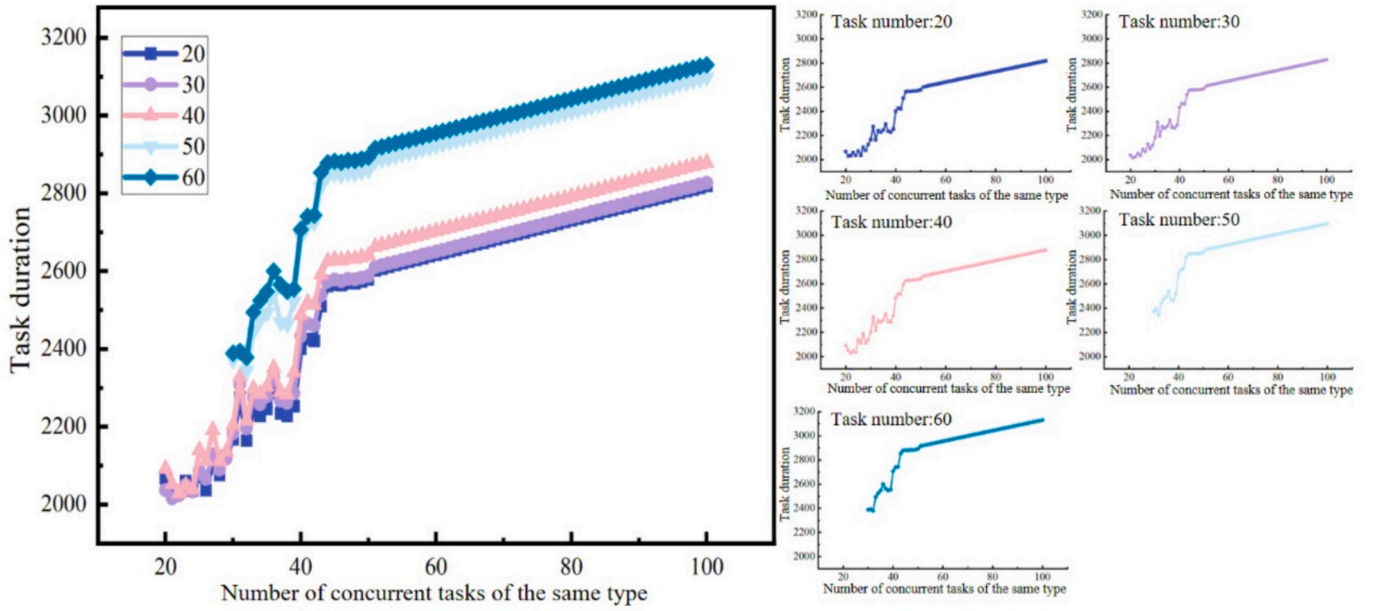


Fig. 10. Task durations for varying numbers of ACT tasks.

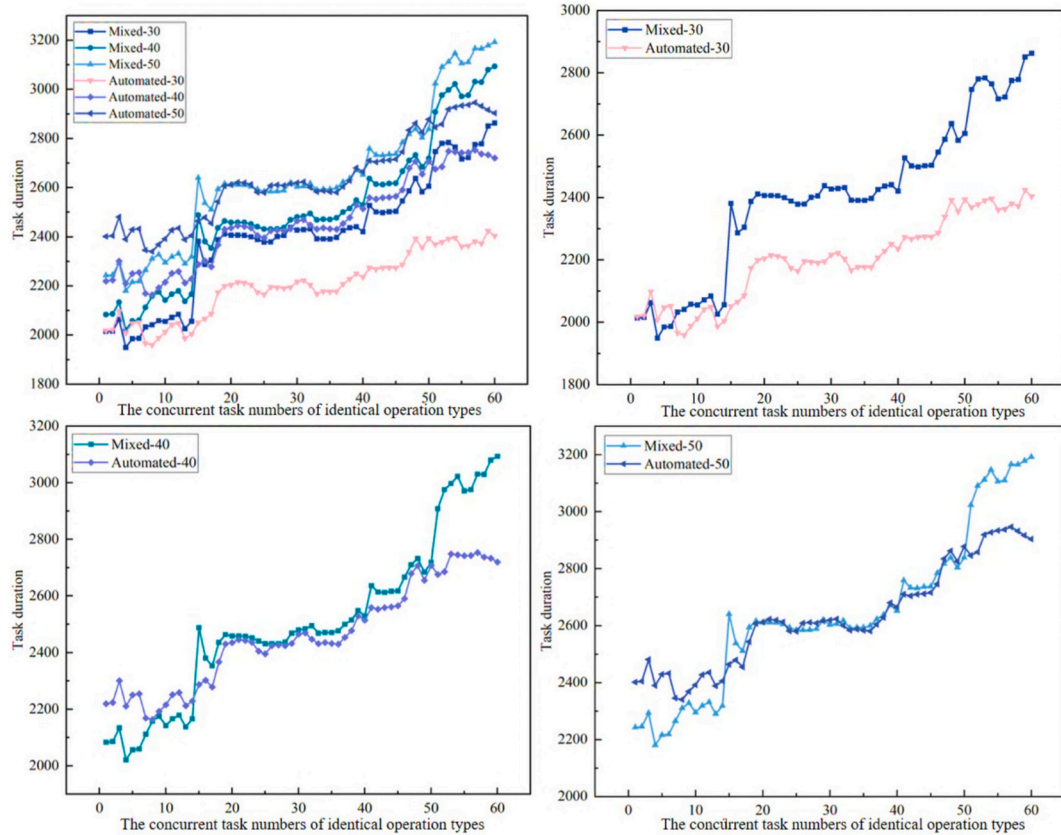


Fig. 11. Task durations under mixed versus fully automated modes.

differences in model predictions with and without each feature, and averaging these differences.

The background dataset, which may be a random subset of the original dataset or the entire dataset, establishes a baseline for assessing feature contribution across various scenarios.

Fig. 14 presents the Shapley value distributions for four machine

learning methods. Features on the y-axis are ordered by importance, with the most influential at the top. Shapley values on the x-axis indicate feature impact on predictions; positive values increase the predicted outcome, whereas negative values decrease it. Each dot corresponds to a single data instance, with red indicating high feature values and blue indicating low feature values.

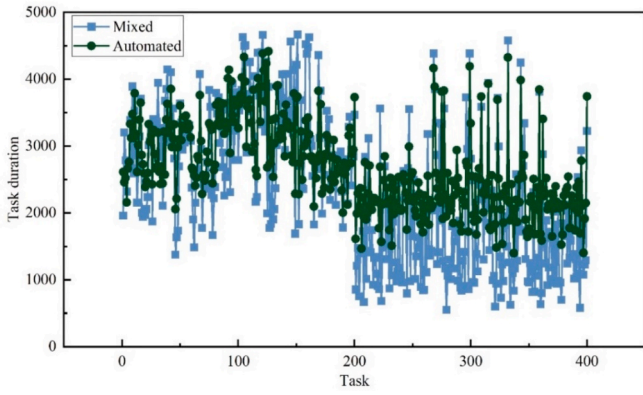


Fig. 12. Task duration under mixed and fully automated modes.

The results demonstrate notable differences in feature importance between neural networks and the three tree-based methods. This divergence likely arises from the fundamental structural differences between neural networks and tree-based algorithms. Notably, specific features, such as F10, exhibit distinct Shapley value interpretations across models. In the FFNN, higher F10 values yield increased outputs, aligning with the positive correlation between F10 and task duration, indicating greater consistency in the FFNN's SHAP results with the correlation analysis. The inconsistencies in other models may stem from feature redundancy. Additionally, features F2, F4, and F9 display minor differences between the neural network and tree-based methods.

Overall, considering both statistical analysis and operational reality, SHAP results appear reasonable. Features F1, F10, F11, and F12 significantly influence task duration, while F6 and F9 have less impact. These results further validate our predictive models, quantitatively illustrating how specific operational variables—such as concurrent task

volume, mixed-truck interaction intensity, and task composition—directly affect terminal efficiency. These results have important academic implications, explicitly guiding future theoretical research by identifying which variables and interactions must be modeled precisely in mixed-mode optimization frameworks and simulations. By clearly quantifying each feature's influence, our SHAP results offer researchers precise targets for in-depth exploration, facilitating the development of advanced modeling techniques capable of capturing complex operational dynamics observed empirically.

Specifically, F10, F11, and F12 reflect operational busyness. F10 represents overall terminal traffic, including queue times at intersections and cranes; F11 indicates the impact of mixed truck numbers on efficiency; and F12 specifically captures congestion and waiting times on roads with uniform directional flow. Higher concurrent task numbers, the number of concurrent tasks operated by different types of trucks, and the number of concurrent tasks of the same type consistently correlate with longer task durations.

Higher F1 values correspond to increased task durations. Here, F1 encodes truck type: “0” for MCT and “1” for ACT. This confirms that MCTs typically perform more efficiently, requiring less time than ACTs due to ACTs' lower speeds and frequent stops for safety. Increased F3 values also correlate with longer task durations. F3 encodes container type: “0” for 40-foot, “1” for 20-foot, and “2” for double 20-foot containers. Tasks involving double 20-foot containers usually require additional time due to differing target locations, often in separate yard blocks.

Features F5, F6, and F7 represent specialized operations with minimal impact. Feature F6 denotes container exchange (0 = no exchange, 1 = failure, 2 = success), showing inconsistent influence. Although successful exchanges might reduce operational delays, failures cause significant waiting and travel inefficiencies, complicating its net effect. Similarly, F5 and F7, indicating human intervention and container relocation respectively, slightly increase waiting times.

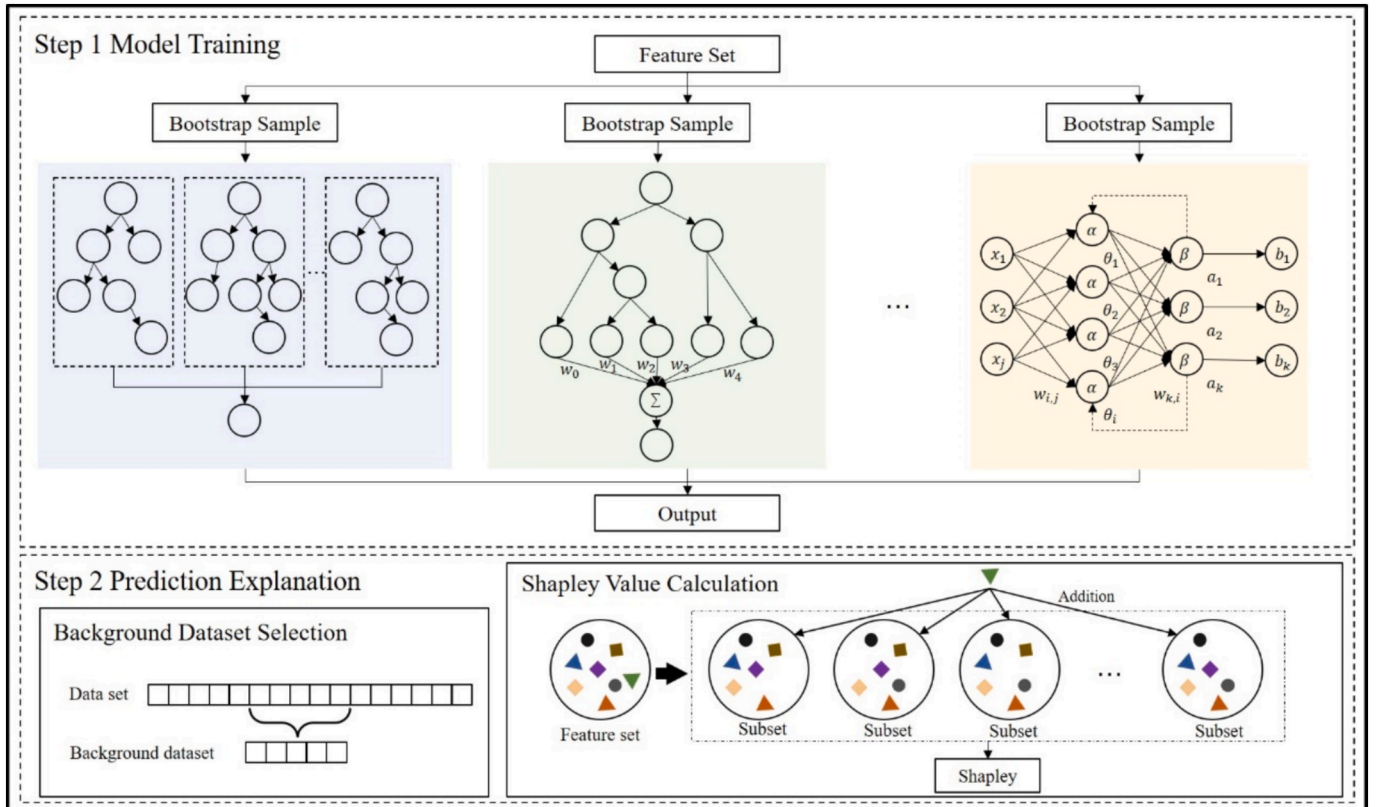


Fig. 13. Shapley value calculation procedure.

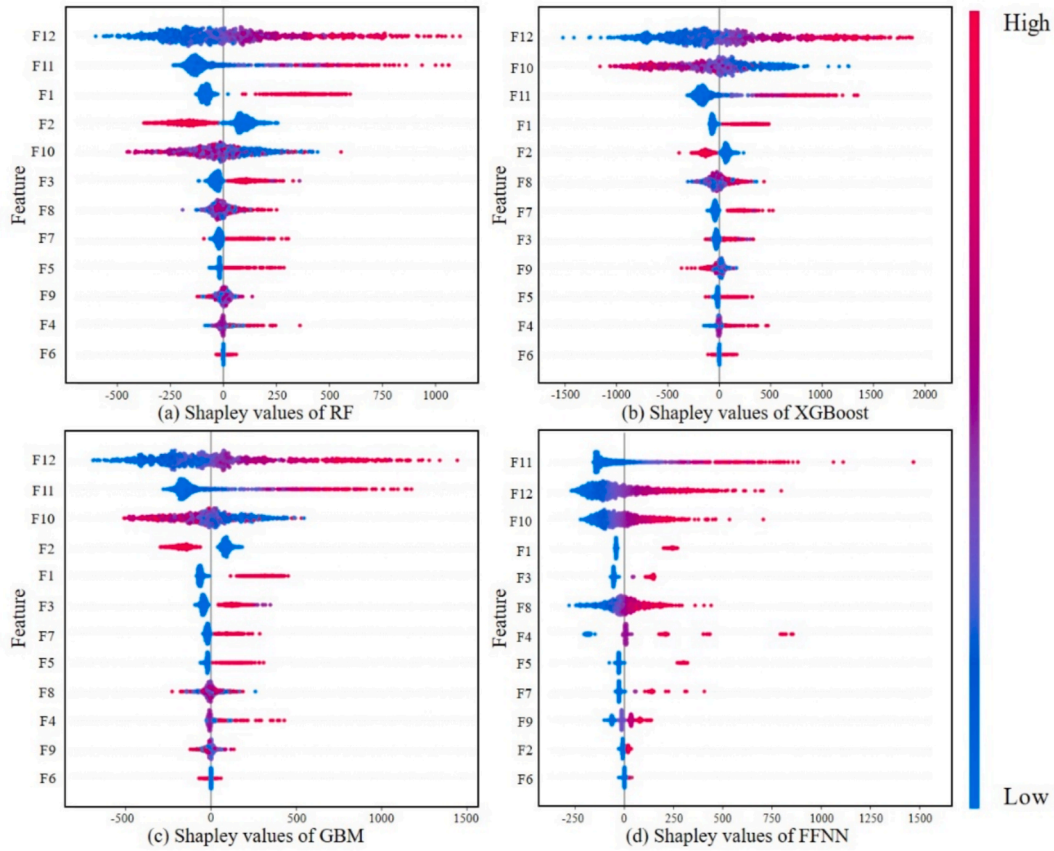


Fig. 14. Shapley values across different machine learning methods.

Container relocation exclusively occurs in loading tasks due to stacking constraints during unloading, which helps explain why loading operations take longer on average. Feature F2, encoding task type (0 = loading, 1 = unloading), further supports this observation. In our dataset, loading tasks exhibit an average total duration of 1,757.7 s and

cover an average distance of 2,502 m, whereas unloading tasks average 1,396.5 s in duration and 1,558 m in distance. These figures confirm that loading tasks incur additional waiting and handling time with more travel, resulting in substantially longer cycle times compared to unloading.

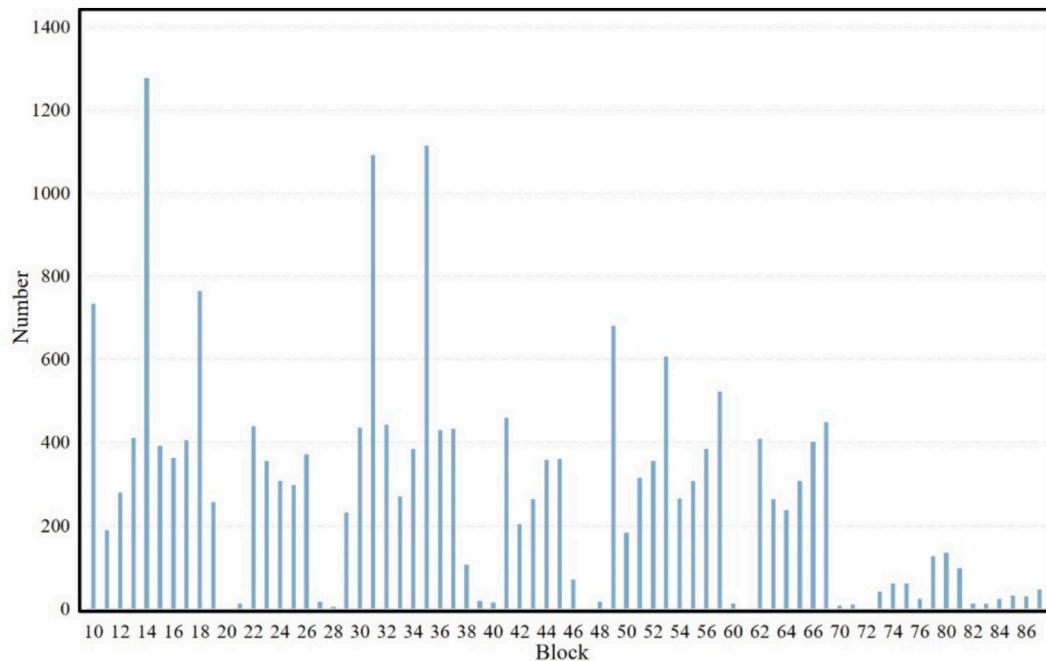


Fig. 15. Distribution of containers across terminal yard blocks.

Finally, F8 and F9 reflect road conditions. The neural network model indicates larger F8 and F9 values lead to increased task duration, although no clear trend emerges in other methods. Given their weak correlation with overall task duration and random container placements, this discrepancy is acceptable. Fig. 15 demonstrates that most tasks occur near the berth in blocks 10–19, 30–39, and 50–57, potentially mitigating the impact of distance and signal delays.

The SHAP analysis highlights critical operational variables influencing task durations: (i) higher numbers of concurrent tasks (features F10–F12) significantly correlate with increased congestion and delays; and (ii) balancing loading and unloading task ratios and maintaining concurrent tasks below identified thresholds (approximately 40 during off-peak periods) substantially improves operational flow. Guided by such findings, terminal managers can proactively schedule interleaved loading/unloading operations and enforce real-time task-volume thresholds to mitigate congestion. Additionally, tasks involving double 20-foot containers (feature F3) typically entail longer durations due to complex routing between multiple yard blocks. Managers can strategically assign such tasks to minimize cross-block movements, allocating ACTs preferentially to handle more complex routing efficiently, leveraging their superior path precision and collision avoidance capabilities.

Given our SHAP results, future research could develop advanced theoretical approaches—such as multi-objective optimization or agent-based simulations—explicitly incorporating identified critical operational thresholds (e.g., limiting concurrent tasks below critical values) and operational policies (e.g., interleaved scheduling, targeted resource allocation strategies). These theoretical advancements, grounded explicitly in empirically-validated operational insights from our study, could substantially enhance the accuracy, realism, and practical applicability of future optimization and simulation models addressing mixed operation environments.

6. Conclusion

This study provides a data-driven analysis of the mixed operation mode in retrofitted container terminals, yielding several critical insights and practical recommendations. Firstly, the current mixed mode is particularly suitable for terminals handling relatively lower daily task volumes. Under these conditions, enhancing the operational priority and maximum speed of ACTs can alleviate the interference from MCTs. Consequently, the full potential of the mixed operation mode can be realized during peak operational periods as automation technology continues to advance. Secondly, the efficiency of container trucks is significantly influenced by the total number of concurrent tasks. An imbalance in the types of tasks, especially during periods of high operational intensity, frequently results in road congestion and increased task durations. Thirdly, loading tasks generally demand more time compared to unloading tasks due to additional requirements for container relocation and longer travel distances. Nevertheless, when the origins and handling procedures of loading and unloading tasks are the same, the durations are comparable. Terminals can improve operational efficiency by allocating additional time specifically for loading tasks and by minimizing unnecessary operations and waiting times.

Due to data constraints and some realistic considerations regarding commercial information, factors such as fixed equipment costs, maintenance, labor expenses, and other managerial concerns are not included in this study. Despite inherent variability across container terminals in terms of precise operational practices and detailed configurations, our study provides clear and empirically supported evidence of core factors significantly influencing mixed-mode operational efficiency. By transparently outlining our operational environment, we enable practitioners and researchers to adapt and extend our empirical findings to conceptually similar scenarios. Thus, our detailed operational characterization and empirical insights collectively ensure robust conceptual generalizability, offering substantial practical value and foundational insights for

subsequent research aimed at formulating generalized optimization methodologies applicable to diverse mixed-mode operational environments.

The results presented in this study are directly derived from real-world operational data, thereby clearly identifying influential factors impacting efficiency under mixed-mode truck operations. It is important to clarify that our analysis is intentionally empirical, focusing on identifying key determinants of operational efficiency under realistic mixed-mode conditions observed within actual terminal operations, which principally highlight immediate practical directions for operational improvement under current conditions rather than explicitly optimal strategies. Although explicit modeling of complex operational constraints (e.g., detailed queue handling logic, collision avoidance mechanisms, precise bottleneck simulations) lies outside our current scope, our data-driven insights provide foundational empirical inputs essential for future simulation-based optimization studies explicitly incorporating these operational constraints, substantially improving the accuracy, practical relevance, and operational validity of proposed optimization frameworks.

Future research on mixed operational modes in container terminals would benefit from employing advanced machine learning techniques and novel analytical methodologies to deepen the understanding of operational efficiency and identify further optimization opportunities. Moreover, comprehensive investigations into additional influential factors—such as integrating various truck types and managing unforeseen disruptions—may provide valuable insights to further enhance terminal efficiency and reliability.

CRedit authorship contribution statement

Chao Lei: Writing – original draft, Validation, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization. **Jia-cheng Liu:** Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Data curation. **Mei Sha:** Writing – review & editing, Supervision, Project administration, Funding acquisition, Conceptualization. **Mei Yang:** Writing – review & editing, Supervision, Resources, Project administration, Data curation, Conceptualization. **Qi Wang:** Formal analysis, Data curation. **Zhongdai Wu:** Writing – review & editing, Resources. **Bing Han:** Writing – review & editing, Resources.

Acknowledgments

This research was supported by the National Natural Science Foundation of China under Grant 72471138, 71972128, 72001136, and the Fund from the State Key Laboratory of Maritime Technology and Safety.

Data availability

The data that has been used is confidential.

References

- Abreu, L. R., Maciel, I. S. F., Alves, J. S., Braga, L. C., & Pontes, H. L. J. (2023). A decision tree model for the prediction of the stay time of ships in Brazilian ports. *Engineering Applications of Artificial Intelligence*, 117, Article 105634. <https://doi.org/10.1016/j.engappai.2022.105634>
- Adi, T. N., Iskandar, Y. A., & Bae, H. (2020). Interterminal truck routing optimization using deep reinforcement learning. *Sensors*, 20, 5794. <https://doi.org/10.3390/s20205794>
- Alvarellos, A., Figuero, A., Carro, H., Costas, R., Sande, J., Guerra, A., Peña, E., & Rabuñal, J. (2021). Machine learning based moored ship movement prediction. *Journal of Marine Science and Engineering*, 9, 800. <https://doi.org/10.3390/jmse9080800>
- Balster, A., Hansen, O., Friedrich, H., & Ludwig, A. (2020). An ETA prediction model for intermodal transport networks based on machine learning. *Business & Information Systems Engineering*, 62, 403–416. <https://doi.org/10.1007/s12599-020-00653-0>
- Breiman, L. (1996). Bagging predictors. *Machine Learning*, 24, 123–140. <https://doi.org/10.1007/BF00058655>

- Container News Team. (2022). Port of Xiamen starts intelligent transformation. *Container News*. <https://container-news.com/port-of-xiamen-starts-intelligent-transformation/>.
- Cuong, T. N., Kim, H.-S., You, S.-S., & Nguyen, D. A. (2022). Seaport throughput forecasting and post COVID-19 recovery policy by using effective decision-making strategy: A case study of Vietnam ports. *Computers & Industrial Engineering*, 168, Article 108102. <https://doi.org/10.1016/j.cie.2022.108102>
- Du, P., Wang, J., Yang, W., & Niu, T. (2019). Container throughput forecasting using a novel hybrid learning method with error correction strategy. *Knowledge-Based Systems*, 182, Article 104853. <https://doi.org/10.1016/j.knosys.2019.07.024>
- Fabregat, A., Vázquez, L., & Vernet, A. (2021). Using machine learning to estimate the impact of ports and cruise ship traffic on urban air quality: The case of Barcelona. *Environmental Modelling & Software*, 139, Article 104995. <https://doi.org/10.1016/j.envsoft.2021.104995>
- Filom, S., Amiri, A. M., & Razavi, S. (2022). Applications of machine learning methods in port operations – a systematic literature review. *Transportation Research Part E: Logistics and Transportation Review*, 161, Article 102722. <https://doi.org/10.1016/j.tre.2022.102722>
- Gao, Y., Chang, D., Fang, T., & Fan, Y. (2019). The daily container volumes prediction of storage yard in port with long short-term memory recurrent neural network. *Journal of Advanced Transportation*, 2019, 1–11. <https://doi.org/10.1155/2019/5764602>
- Jiang, X., & Shang, Q. (2023). A dynamic CAV-dedicated lane allocation method with the joint optimization of signal timing parameters and smooth trajectory in a mixed traffic environment. *IEEE Transactions on Intelligent Transportation Systems*, 24(6), 6436–6449. <https://doi.org/10.1109/TITS.2022.3172942>
- Kim, J. H., Kim, J., Lee, G., & Park, J. (2021). Machine learning-based models for accident prediction at a Korean container port. *Sustainability*, 13, 9137. <https://doi.org/10.3390/su13169137>
- Knatz, G., Notteboom, T., & Pallis, A. A. (2022). Container terminal automation: Revealing distinctive terminal characteristics and operating parameters. *Maritime Economics & Logistics*, 24(3), 537–565. <https://doi.org/10.1057/s41278-022-00240-y>
- Kourounioti, I., Polydoropoulou, A., & Tsiklidis, C. (2016). Development of models predicting dwell time of import containers in port container terminals – an artificial neural networks application. *Transportation Research Procedia*, 14, 243–252. <https://doi.org/10.1016/j.trpro.2016.05.061>
- Li, J., Zhang, S., & Xu, B. (2024). Containership delay propagation risk analysis based on effective multivariate transfer entropy. *Ocean Engineering*, 298, Article 117077. <https://doi.org/10.1016/j.oceaneng.2024.117077>
- Liu, Q., Zhang, K., Li, M., Chen, X., Lin, X., & Li, S. (2024). Integrated optimization of traffic signal timings and vehicle trajectories considering mandatory lane-changing at isolated intersections. *Transportation Research Part C: Emerging Technologies*, 163, Article 104614. <https://doi.org/10.1016/j.trc.2024.104614>
- Lundberg, S. M., & Lee, S.-I. (2017). A unified approach to interpreting model predictions (pp. 4768–4777). NY, USA: Red Hook. <https://doi.org/10.5555/3295222.3295230>
- Luo, Y., & Huang, L. (2020). Port short-term truck flow forecasting model based on wavelet neural network. In *2020 International Conference on Intelligent Transportation, Big Data & Smart City (ICITBS)* (pp. 43–47). Vientiane, Laos: IEEE. <https://doi.org/10.1109/ICITBS49701.2020.00017>
- Ma, W., Li, J., & Yu, C. (2022). Shared-phase-dedicated-lane based intersection control with mixed traffic of human-driven vehicles and connected and automated vehicles. *Transportation Research Part C: Emerging Technologies*, 135, Article 103509. <https://doi.org/10.1016/j.trc.2021.103509>
- Milenković, M., Milosavljević, N., Bojović, N., & Val, S. (2021). Container flow forecasting through neural networks based on metaheuristics. *Operational Research*, 21, 965–997. <https://doi.org/10.1007/s12351-019-00477-1>
- Niroumand, R., Hajibabai, L., & Hajbabaie, A. (2023). White phase intersection control through distributed coordination: A mobile controller paradigm in a mixed traffic stream. *IEEE Transactions on Intelligent Transportation Systems*, 24(3), 2993–3007. <https://doi.org/10.1109/TITS.2022.3226557>
- Oliveira, R. M. A., Sant'Anna, A. M. O., & Da Silva, P. H. F. (2024). Explainable machine learning models for defects detection in industrial processes. *Computers & Industrial Engineering*, 192, Article 110214. <https://doi.org/10.1016/j.cie.2024.110214>
- Ozturk, U., Birbil, S. I., & Cicek, K. (2019). Evaluating navigational risk of port approach manoeuvres with expert assessments and machine learning. *Ocean Engineering*, 192, Article 106558. <https://doi.org/10.1016/j.oceaneng.2019.106558>
- Pani, C., Vanelslander, T., Fancello, G., & Cannas, M. (2015). Prediction of late/early arrivals in container terminals – a qualitative approach. *European Journal of Transport and Infrastructure Research*, 15. <https://doi.org/10.18757/ejtr.2015.15.4.3096>
- Ping, F. F., & Fei, F. X. (2013). Multivariate Forecasting Mode of Guangdong Province Port throughput with Genetic Algorithms and Back Propagation Neural Network. *Procedia - Social and Behavioral Sciences*, 96, 1165–1174. <https://doi.org/10.1016/j.sbspro.2013.08.133>
- Santos, J. I., Pereda, M., Ahedo, V., & Galán, J. M. (2023). Explainable machine learning for project management control. *Computers & Industrial Engineering*, 180, Article 109261. <https://doi.org/10.1016/j.cie.2023.109261>
- Skaf, A., Lamrous, S., Hammoudan, Z., & Manier, M.-A. (2021). Integrated quay crane and yard truck scheduling problem at port of Tripoli-Lebanon. *Computers & Industrial Engineering*, 159, Article 107448. <https://doi.org/10.1016/j.cie.2021.107448>
- Song, X., Chen, N., Zhao, M., Wu, Q., Liao, Q., & Ye, J. (2024). Novel AGV resilient scheduling for automated container terminals considering charging strategy. *Ocean & Coastal Management*, 250, Article 107014. <https://doi.org/10.1016/j.ocecoaman.2023.107014>
- Stepce, D., Martincic, T., Klein, F., Vladusic, D., & Costa, J. P. (2020). Machine learning based system for vessel turnaround time prediction. In *2020 21st IEEE International Conference on Mobile Data Management (MDM)* (pp. 258–263). <https://doi.org/10.1109/MDM48529.2020.00060>
- Tajalli, M., & Hajbabaie, A. (2022). Traffic signal timing and trajectory optimization in a mixed autonomy traffic stream. *IEEE Transactions on Intelligent Transportation Systems*, 23(7), 6525–6538. <https://doi.org/10.1109/TITS.2021.3058193>
- Tao, Y., Zhang, S., Lin, C., & Lai, X. (2023). A bi-objective optimization for integrated truck operation and storage allocation considering traffic congestion in container terminals. *Ocean & Coastal Management*, 232, Article 106417. <https://doi.org/10.1016/j.ocecoaman.2022.106417>
- Gökkuğ, Ü., Yildırım, M. S., & Aydin, M. M. (2017). Estimation of container traffic at seaports by using several soft computing methods: A case of Turkish seaports. *Discrete Dynamics in Nature and Society*, 2017, 1–15. <https://doi.org/10.1155/2017/2984853>
- Wang, Q., Gong, Y., & Yang, X. (2022). Connected automated vehicle trajectory optimization along signalized arterial: A decentralized approach under mixed traffic environment. *Transportation Research Part C: Emerging Technologies*, 145, Article 103918. <https://doi.org/10.1016/j.trc.2022.103918>
- Wu, M., Gao, J., Li, L., & Wang, Y. (2022). Control optimisation of automated guided vehicles in container terminal based on Petri network and dynamic path planning. *Computers & Electrical Engineering*, 104, Q2. <https://doi.org/10.1016/j.compeleceng.2022.108471>
- Xie, Y., Chen, Y., Wei, Q., & Yin, H. (2024). A hybrid deep learning approach to improve real-time effluent quality prediction in wastewater treatment plant. *Water Research*, 250, Article 121092. <https://doi.org/10.1016/j.watres.2023.121092>
- Xie, Y., & Huynh, N. (2010). Kernel-based machine learning models for predicting daily truck volume at seaport terminals. *Journal of Transportation Engineering*, 136, 1145–1152. [https://doi.org/10.1061/\(ASCE\)TE.1943-5436.0000186](https://doi.org/10.1061/(ASCE)TE.1943-5436.0000186)
- Xu, B., Fan, H., & Li, J. (2024). Energy-efficiency and safety-driven multidimensional evolutionary game for AGVs transportation at automated container terminals. *Computers & Industrial Engineering*, 192, Article 110192. <https://doi.org/10.1016/j.cie.2024.110192>
- Xu, B., Jie, D., Li, J., Yang, Y., Wen, F., & Song, H. (2021). Integrated scheduling optimization of U-shaped automated container terminal under loading and unloading mode. *Computers & Industrial Engineering*, 162, Article 107695. <https://doi.org/10.1016/j.cie.2021.107695>
- Xu, J., Chen, T., Fang, X., Xia, L., & Pan, X. (2024). Prediction model of pressure injury occurrence in diabetic patients during ICU hospitalization—XGBoost machine learning model can be interpreted based on SHAP. *Intensive and Critical Care Nursing*, 83, Article 103715. <https://doi.org/10.1016/j.iccn.2024.103715>
- Yang, C., Guan, X., Xu, Q., Xing, W., Chen, X., Chen, J., & Jia, P. (2024). How can SHAP (SHapley Additive exPlanations) interpretations improve deep learning based urban cellular automata model? *Computers Environment and Urban Systems*, 111, Article 102133. <https://doi.org/10.1016/j.compenvurbysys.2024.102133>
- Yang, H., Ma, W., Cheng, R., Wu, B., Wang, Y., & Zheng, P. (2025). A novel game approach to integrating traffic assignment and signal control for enhanced efficiency and environmental performance in mixed networks. *Transportation Research Part C: Emerging Technologies*, 171, Article 105017. <https://doi.org/10.1016/j.trc.2025.105017>
- Yu, J., Tang, G., Song, X., Yu, X., Qi, Y., Li, D., & Zhang, Y. (2018). Ship arrival prediction and its value on daily container terminal operation. *Ocean Engineering*, 157, 73–86. <https://doi.org/10.1016/j.oceaneng.2018.03.038>
- Zhang, J., Chang, C., Li, S., Ban (Jeff), X., & Li, L. (2024). Unleashing the two-dimensional benefits of connected and automated vehicles via dedicated intersections in mixed traffic. *Transportation Research Part C: Emerging Technologies*, 160, Article 104501. <https://doi.org/10.1016/j.trc.2024.104501>